

Article

Real-Time 2D Orthomosaic Mapping from UAV Video via Feature-Based Image Registration

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Abstract

This study presents a real-time framework for generating two-dimensional (2D) orthomosaic maps directly from UAV video. The method targets operational scenarios in which a continuously updated 2D overview is required during flight or immediately after landing, without relying on time-consuming offline photogrammetry workflows such as structure-from-motion (SfM) and multi-view stereo (MVS). The proposed procedure incrementally registers sparsely sampled video frames on standard CPU hardware using classical feature-based image registration. Each selected frame is converted to grayscale and processed under a fixed keypoint budget to maintain predictable runtime. Tentative correspondences are obtained through descriptor matching with ratio-test filtering, and outliers are removed using random sample consensus (RANSAC) to ensure geometric consistency. Inter-frame motion is modeled by a planar homography, enabling the mapping process to jointly account for rotation, scale variation, skew, and translation that commonly occur in UAV video due to yaw maneuvers, mild altitude variation, and platform motion. Sequential homographies are accumulated to warp incoming frames into a global mosaic canvas, which is updated incrementally using lightweight blending suitable for real-time visualization. Experimental results on three UAV video sequences with different durations, flight patterns, and scene targets report representative orthomosaic-style outputs and per-step CPU runtime statistics (mean, 95th percentile, and maximum), illustrating typical operating behavior under the tested settings. The framework produces visually coherent orthomosaic-style maps in real time for approximately planar scenes with sufficient overlap and texture, while clarifying practical failure modes under weak texture, motion blur, and strong parallax. Limitations include potential drift over long sequences and the absence of ground-truth references for absolute registration-error evaluation.

Keywords: real-time orthomosaic; incremental mosaicking; feature-based registration; planar homography; RANSAC



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1. Introduction

Unmanned aerial vehicles (UAVs) have become widely adopted platforms for acquiring high-resolution imagery across precision agriculture, environmental monitoring, infrastructure inspection, and disaster response [1–6]. Low-altitude flights enable centimeter-level ground sampling distance and flexible revisit cycles, making UAV imagery attractive for time-sensitive field operations compared with conventional airborne campaigns [2–6]. In

many practical scenarios, however, individual frames are insufficient; an integrated two-dimensional (2D) orthomosaic is preferred because it supports rapid situational awareness, navigation support, and change interpretation over a wide area [2,3,6,7]. Typical examples include rapid post-disaster inspection, road monitoring for traffic incidents, infrastructure inspection over extended areas (e.g., roofs, bridges, roads, and construction sites), and site-scale situational awareness for search-and-rescue or security patrols, where a single frame provides only a limited field of view and an integrated 2D overview enables wide-area interpretation and navigation support. In time-critical missions, it is desirable to update such a 2D map during flight or immediately after landing using field-deployable computing resources with predictable latency.

Orthomosaic generation is commonly performed using photogrammetric pipelines based on structure-from-motion (SfM) and multi-view stereo (MVS) [1–3,8]. Commercial and open-source toolchains implement these workflows and provide mature documentation, including examples of end-to-end processing and reproducibility guidance [9,10]. Despite their accuracy potential, SfM/MVS pipelines remain computationally demanding and are typically executed offline. They require dense image collections with sufficient overlap, and processing can take minutes to hours depending on dataset size, image resolution, and reconstruction settings [1–3,8–11]. These characteristics limit operational usefulness when an updated map is needed during flight, immediately after landing, or under bandwidth and computing constraints.

To address this gap, a growing body of work has investigated real-time or near-real-time mosaicking and registration frameworks that operate directly on image sequences [12–25]. These approaches generally rely on sequential processing, where local features are extracted, matched, and used to estimate inter-frame motion, followed by incremental warping and blending to update a mosaic [14,15,17,20–22]. Several studies report that image-only pipelines can reduce system complexity compared with solutions that rely on external navigation sensors, while still enabling practical mapping outputs for monitoring and decision support [15,16,26–33]. For such incremental pipelines, practical performance is largely governed by update latency and runtime predictability at each registration-and-update step.

A critical design choice in incremental mosaicking is the transformation model used to relate frames. Simplified motion models such as translation or similarity transforms can be efficient, but they can be inadequate for UAV video because yaw maneuvers, mild altitude variation, and platform motion introduce rotation, scale change, and skew on the image plane [6,21,28]. Planar homography provides a unified projective model that can represent translation, rotation, anisotropic scaling, shear, and mild perspective effects, and it is widely used in panoramic stitching and aerial registration [34]. Robust estimation strategies such as random sample consensus (RANSAC) are typically required because outliers are common in feature correspondences [35,36]. In addition, practical pipelines must balance robustness and runtime, which motivates careful control of keypoint budgets, matching strategies, and parameter choices [37–43].

Another practical challenge is cumulative error. Incremental composition of inter-frame transforms can accumulate drift over long sequences, which may degrade alignment and introduce distortions in the mosaic [20–22]. Drift mitigation can be addressed through global optimization, loop closure, or sensor fusion, but these additions increase system complexity and often require stronger compute resources [16–19]. Moreover, as sequences grow, real-time systems can encounter scalability issues due to expanding canvas size, increased memory and blending cost, and accumulated drift; therefore, practical pipelines often balance update frequency, transformation complexity, and blending strategy to maintain throughput and map usability [33,44–47]. In this context, a pragmatic operating

regime for time-critical field applications is to emphasize stable short-horizon registration and consistent update speed, while explicitly modeling rotation, scale change, skew, and translation with a full homography model. Accordingly, the experiments in this paper are designed to characterize representative operating behavior and CPU runtime characteristics under controlled compute budgets, rather than to establish broad statistical generalization across large-scale benchmarks.

This paper targets an operating regime that prioritizes real-time 2D orthomosaic-style updates from UAV video under CPU-oriented constraints, using classical feature-based image registration and a planar homography motion model without relying on heavy 3D reconstruction. Accordingly, the contribution of this work is system-oriented: rather than proposing a new feature detector, a new descriptor, or a new robust estimator, we emphasize real-time design choices (frame sampling and bounded keypoint processing) to achieve predictable runtime and practical mapping behavior in UAV video mosaicking.

The main contributions are as follows:

- A CPU-feasible real-time incremental 2D orthomosaic-style mapping framework that operates directly on UAV video via explicit frame sampling and bounded keypoint processing, enabling continuous map updates without offline SfM/MVS.
- A system-level design for predictable runtime and stable updates, including fixed processing budgets and lightweight mosaic update logic suited to real-time visualization on standard CPUs.
- Explicit adoption and accumulation of a full planar homography model to capture inter-frame transformations typical of UAV motion, together with a clear discussion of operating conditions and practical limitations.
- A controlled, reproducible evaluation protocol, including an ablation-style analysis, that quantifies the effects of key design parameters on mapping quality and runtime, providing practical configuration guidelines for deployment.

Compared with offline SfM/MVS workflows that aim for globally consistent reconstruction, the present work emphasizes real-time CPU operation for incremental mosaicking from nadir UAV video by enforcing explicit compute budgets (frame sampling and bounded keypoint processing), while adopting a full planar homography formulation to model inter-frame deformation beyond translation-only alignment. We explicitly limit the claims to this CPU-feasible real-time operating regime and discuss the need for larger-scale and more diverse benchmarking as future work.

The remainder of this paper is organized as follows. Section 2 reviews related work in UAV orthomosaic generation, real-time incremental mosaicking, feature-based registration, and transformation modeling, and it clarifies the positioning of the present study within these research streams. Section 3 describes the proposed processing procedure, including frame extraction, preprocessing, feature detection and matching, robust homography estimation, and incremental orthomosaic construction. Section 4 details the experimental setup, datasets, parameter settings, and baseline configurations. Section 5 presents qualitative and quantitative results, including orthomosaic examples and runtime characteristics. Finally, the paper summarizes the main findings and practical limitations relevant to real-world deployment.

2. Related Work

2.1. UAV Orthomosaic Generation via SfM/MVS

Orthomosaic generation from UAV imagery is most commonly performed using photogrammetric pipelines based on structure-from-motion (SfM) and multi-view stereo (MVS) [1–3,8]. Commercial and open-source toolchains provide mature workflows and documentation, including end-to-end processing examples and reproducibility guidance [9–11].

Despite their potential for high-quality and globally consistent reconstructions, SfM/MVS pipelines are typically executed offline and remain computationally demanding, requiring dense image collections with sufficient overlap and careful parameter settings [1–3,8–11]. Processing time often ranges from minutes to hours, depending on dataset size, image resolution, and reconstruction configuration [9–11]. These characteristics limit operational usefulness when a continuously updated 2D overview is required during flight or immediately after landing under bandwidth and computing constraints.

2.2. Real-Time and Incremental Mosaicking from Image Sequences

To reduce latency and system complexity, many studies have investigated real-time or near-real-time mosaicking and registration frameworks that operate directly on image sequences [12–15,31]. These approaches typically follow a sequential processing structure, where local features are extracted and matched to estimate inter-frame motion, followed by incremental warping and mosaic updates [14,15,20]. Image-only pipelines can be attractive in field operations because they avoid dependence on external navigation sensors while still enabling practical mapping outputs for monitoring and decision support [15,16,31]. However, incremental mosaicking faces scalability issues as sequences grow. Computational cost can increase as the mosaic canvas expands (e.g., larger warping and blending regions and less favorable memory access), and sequential transform composition can introduce drift over long trajectories [14,21,22]. Practical real-time systems, therefore, balance update frequency, transformation complexity, and blending strategy to maintain throughput and map usability, commonly adopting lightweight update logic and conservative parameter choices under fixed compute budgets [20,44–47].

In addition to feature-based sequential pipelines, recent studies have explored application-driven mosaicking/registration variants and stitching-quality improvements to enhance visual coherence under challenging conditions, although such enhancements may introduce extra computation or stronger modeling assumptions [44–47].

2.3. Feature Detection, Description, and Matching for UAV Video Registration

Classical feature-based registration remains widely adopted due to its interpretability and broad availability in CPU implementations [34,37–39,43]. Scale- and rotation-invariant features such as SIFT and SURF have been extensively used for aerial and remote-sensing mosaicking because they provide relatively stable correspondences under viewpoint and scale changes [29,37,38]. Binary descriptors such as ORB are often preferred in real-time settings due to their reduced computation and matching cost, although their robustness can be more sensitive to texture quality and imaging conditions [39]. Feature matching is commonly performed using nearest-neighbor search strategies (e.g., k-nearest neighbors (KNN)), sometimes accelerated with approximate methods such as fast library for approximate nearest neighbors (FLANN), followed by ratio tests or mutual consistency checks to reduce ambiguous correspondences [42]. Robust geometric verification is typically required because UAV imagery often contains repeated structures, motion blur, or low-texture regions that introduce outliers in tentative matches [35].

Beyond classical features, learning-based local features and matching (e.g., SuperPoint/SuperGlue) have been reported to improve correspondence quality in some conditions, but they are often less convenient for strict CPU-only real-time deployment due to additional compute and dependency requirements [40,41]. Deep-feature-based registration has also been explored for UAV remote-sensing imagery, showing potential robustness improvements at the expense of additional model complexity and nontrivial runtime overhead in some deployments [33]. In practical field deployment, the choice of feature and

matching configuration must therefore consider robustness, reproducibility, and compute feasibility under target hardware constraints.

2.4. Transformation Modeling and Stitching Quality Considerations

A key design choice in mosaicking is the transformation model used to relate successive frames. Simplified motion models such as translation or similarity transforms can be computationally efficient, but they may be inadequate for UAV video because yaw maneuvers, altitude variations, and platform motion can introduce rotation, scale change, skew, and mild perspective effects [6,21,28]. Planar homography provides a unified projective model capable of representing translation, rotation, anisotropic scaling, shear, and limited perspective effects, and it has been widely used in panoramic stitching and aerial registration [34]. Robust estimation techniques such as random sample consensus (RANSAC) are commonly adopted to estimate homographies in the presence of outliers [35].

Stitching quality also depends on how warped images are merged into the mosaic. Blending strategies, seam selection, and local refinement can reduce visible seams and ghosting artifacts but may increase computational cost or require additional modeling [12,34]. Recent work has investigated seam-aware or region-aware stitching strategies to improve visual continuity in UAV image stitching [44]. Other studies have explored mesh-guided deformation or constraint-based alignment to better handle local misalignment when planar assumptions are violated, although these approaches typically introduce more complex optimization and may reduce real-time predictability on CPU platforms [45]. Application-driven mosaic variants (e.g., adaptive mosaics for online inspection) also illustrate that improving visual coherence often requires explicit trade-offs among model complexity, update frequency, and computational load [47].

Overall, prior studies indicate that real-time incremental mosaicking systems must make explicit trade-offs among transformation complexity, correspondence robustness, mosaic update logic, and blending strategy to sustain throughput while maintaining usable map quality as sequences grow [15,20,45–47]. This motivates system-level designs that enforce bounded compute budgets and predictable runtime when CPU-only real-time deployment is required.

2.5. Homography-Based Mapping and Limitations

Homography models are widely used in image stitching and can represent combined in-plane rotation, scaling, shear, and translation, with additional projective effects under planar or locally planar assumptions [34]. UAV image sequences may satisfy approximate planarity at low altitude over ground-like surfaces, although violations occur in scenes with strong relief or tall structures [1–3]. A known limitation of incremental homography accumulation is drift, which may appear as misalignment and distortion as sequence length increases [20–22]. Hybrid approaches that improve stability include additional constraints, global optimization, or sensor integration, but these typically increase computational load and system complexity [16,19].

Even when a suitable transformation model is used, long sequences remain vulnerable to cumulative error. Reported practical strategies to improve stability include maintaining sufficient overlap, using robust estimation, and adopting conservative mosaic update logic to preserve throughput as the canvas grows [14,15,20]. These constraints motivate experimental evaluation that considers both orthomosaic usability and runtime feasibility.

2.6. Positioning of the Present Study

Existing UAV mapping solutions span a spectrum from offline SfM/MVS pipelines that provide global consistency and high mapping accuracy [1–3,8] to sequential mosaicking

methods that aim to reduce latency by updating a 2D map incrementally from image streams [12–20]. Within incremental mosaicking, practical performance depends on the transformation model, the robustness of correspondence formation, and the computational budget allocated per update step [34,35,37–39]. Moreover, scalability and visual usability are influenced by mosaic canvas growth and blending/seam strategies, which remain active considerations in UAV stitching and related real-time mosaicking applications [44,45].

Accordingly, this study is positioned as a system-oriented contribution that targets CPU-feasible real-time incremental mosaicking from nadir UAV video under explicit compute constraints. Rather than proposing a new detector, descriptor, or robust estimator, the focus is on predictable runtime behavior via bounded keypoint processing and explicit frame sampling, together with practical evaluation under controlled conditions.

3. Real-Time Orthomosaic Mapping Procedure

3.1. Overview of the Processing Procedure

The proposed mapping procedure constructs a two-dimensional (2D) orthomosaic-style map by incrementally registering sampled frames from a UAV video stream. Figure 1 summarizes the workflow, including frame sampling, preprocessing, feature-based registration, robust homography estimation, homography accumulation, warping to a global canvas, and mosaic blending and update. The target operating regime assumes that the observed ground region is predominantly planar at the working altitude, such that the dominant inter-frame geometry can be approximated by a planar homography.

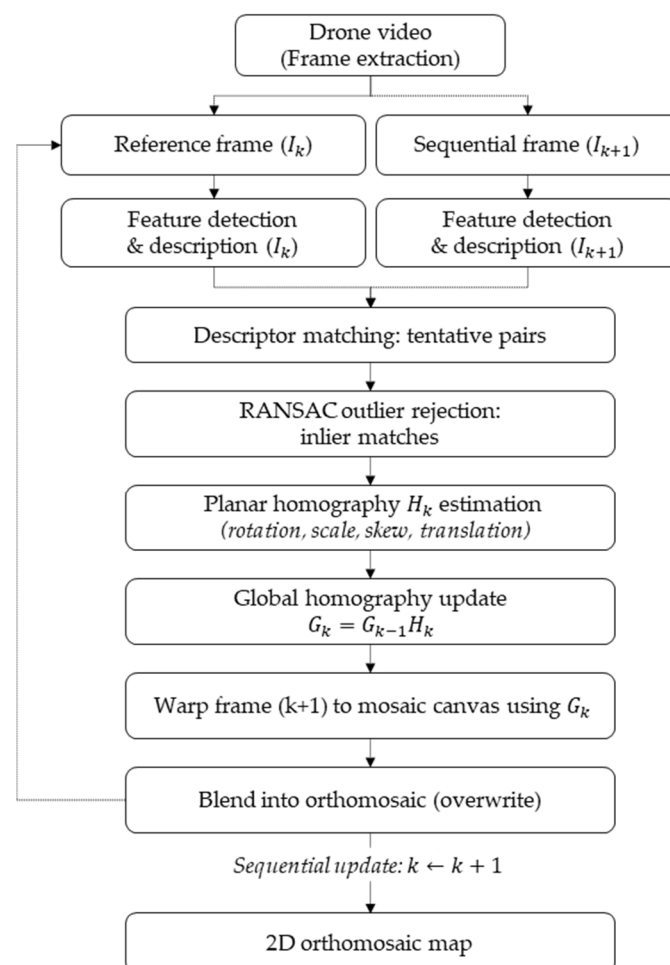


Figure 1. Overall procedure of the proposed real-time 2D orthomosaic mapping from UAV video.

A UAV-mounted RGB camera records a continuous video stream during flight. Instead of processing all frames, the pipeline extracts frames at a fixed sampling interval N_s , meaning that every N_s -th video frame is selected to form the extracted-frame sequence. This design reduces computational load while preserving sufficient overlap between consecutive extracted frames for stable registration, which is essential for CPU-oriented real-time operation. For a video frame rate f (frames/s), the corresponding update period is $T_s = N_s / f$. At update step t , a reference grayscale frame I_k and the next grayscale frame I_{k+1} in the extracted sequence are processed as a pair. Feature detection and description are applied to both frames, and tentative correspondences are obtained through descriptor matching. For predictable runtime on standard CPUs, the number of retained keypoints per frame is bounded by a keypoint budget $N_{\max}$ (e.g., retaining keypoints with the strongest responses). This bounded keypoint processing directly controls the dominant cost of matching and robust estimation.

A planar homography $H_{k \rightarrow k+1}$ is estimated to register I_{k+1} to I_t using random sample consensus (RANSAC)-based outlier rejection. The global mapping transform is then updated by sequentially composing the accepted inter-frame homographies over time, yielding an accumulated transform that maps each extracted frame into a common mosaic-canvas coordinate system. If the estimated model does not satisfy a minimum acceptance criterion (e.g., insufficient inlier support), the update is rejected to avoid propagating an unreliable transform in the accumulated mapping. The accepted frame is then warped onto the global canvas using the accumulated transform, and the mosaic is updated using a lightweight blending rule to maintain real-time throughput.

The purpose of this procedure is not to introduce a new detector, descriptor, or robust estimator, but to provide a CPU-feasible incremental mosaicking workflow whose runtime is controlled through explicit sampling (N_s) and bounded keypoint processing ($N_{\max}$), and whose behavior is evaluated under consistent matching and RANSAC criteria [12–15,20–22]. Consistent with Sections 4 and 5, real-time feasibility is assessed by comparing step-level CPU runtime against the update period T_s .

3.2. Frame Extraction from UAV Video

The input is a UAV video stream, typically recorded at full HD resolution (e.g., 1920×1080 pixels) or comparable formats. Processing every video frame is generally unnecessary and may prevent real-time operation on commodity CPUs. Frames are therefore extracted from the UAV video at a fixed sampling interval N_s , i.e., every N_s -th frame is selected for processing. The extracted frames form a sequential set, and the pipeline performs registration and mosaic updates using consecutive extracted frames.

For a video recorded at a frame rate f_v (frames/s), the update period associated with the extracted frames is defined as Equation (1).

$$T_s = \frac{N_s}{f_v} \quad (1)$$

The procedure is considered feasible for real-time updates when the step-level processing time t_{step} remains consistently below the update period T_s under the chosen configuration. This feasibility condition is used consistently in the runtime analysis reported in Sections 4 and 5.

Each extracted RGB frame is converted to grayscale prior to feature processing to reduce computational load while preserving the intensity structures that dominate classical feature responses. In this paper, the extracted RGB frames are denoted by I_k^{RGB} , and the corresponding grayscale frames used for registration are denoted by I_k . Optional resizing and cropping can be applied when needed to stabilize runtime and suppress

non-ground regions; when used, the same preprocessing rule is applied consistently within each sequence to avoid introducing time-varying distortions.

As illustrated in Figure 2, the feature-extraction stage can be made robust to inter-frame scale variation by using a scale-space approach such as the Difference-of-Gaussians (DoG) pyramid in SIFT, where extrema are detected across both image location and scale. In this work, the sampling interval N_s and the keypoint budget $N_{\max}$ are treated as explicit system-level design parameters to ensure predictable CPU runtime in a real-time operating regime. Although overly aggressive settings may reduce correspondence robustness and registration stability, the objective here is to characterize and communicate this accuracy–latency trade-off under fixed computational constraints, rather than to introduce a new lightweight feature or matching algorithm.

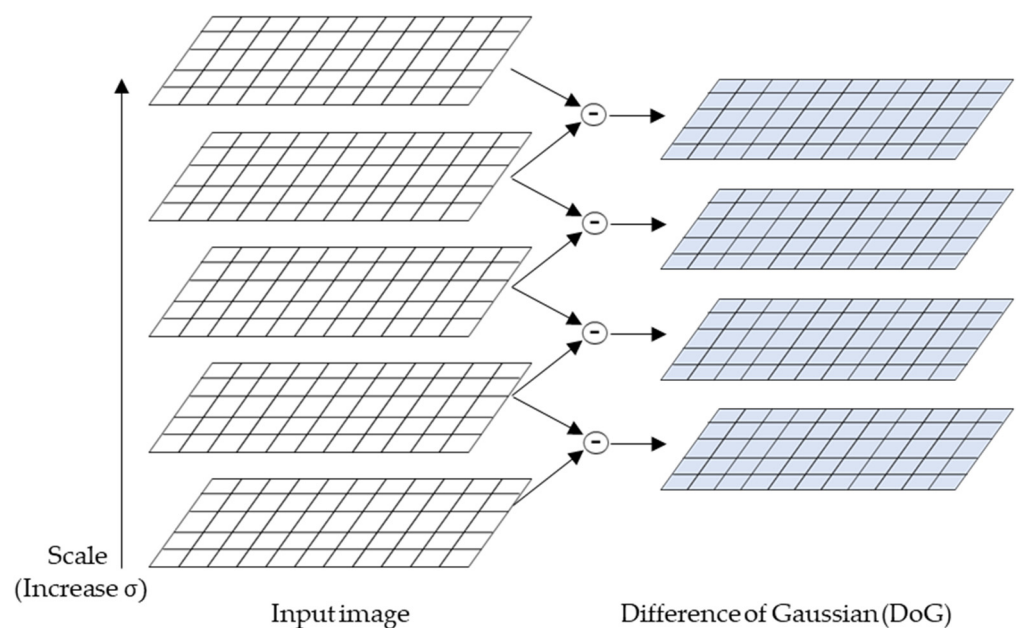


Figure 2. Illustration of SIFT scale-space construction and difference-of-Gaussians (DoG) keypoint detection used for feature extraction.

3.3. Feature Detection and Description

Feature detection and description provide correspondence candidates required for estimating the inter-frame homography. UAV video sequences may exhibit illumination variation, motion blur, and viewpoint changes; therefore, robustness to scale and in-plane rotation can be important for stable registration. Classical local features remain widely used because they are interpretable, deployable without training, and compatible with CPU-based pipelines. Representative detectors and descriptors include SIFT, SURF, and ORB [37–39].

As conceptually illustrated in Figure 2, SIFT builds a scale-space representation to support scale robustness when apparent scale changes occur between consecutive extracted frames. In SIFT, the scale space is defined by convolving the grayscale image I_t with a Gaussian kernel as follows Equation (2).

$$L(\mathbf{x}, \sigma) = G(\mathbf{x}, \sigma) * I(\mathbf{x}) \quad (2)$$

where $G(\mathbf{x}, \sigma)$ denotes the Gaussian kernel, $*$ denotes convolution, and σ is the scale parameter [37]. Keypoints are detected as extrema in the DoG pyramid (Equation (3)).

$$D(\mathbf{x}, \sigma) = L(\mathbf{x}, k\sigma) - L(\mathbf{x}, \sigma) \quad (3)$$

where k is a multiplicative factor between adjacent scales [37]. After detection, an orientation is assigned, and a descriptor vector is computed from local gradient statistics to support matching under in-plane rotation and moderate photometric variation [37].

To preserve real-time throughput and maintain stable homography estimation across extracted frames, detector thresholds and the number of retained keypoints are controlled. Specifically, we impose a keypoint budget by retaining at most $N_{\max}$ keypoints per extracted frame (e.g., selecting keypoints based on the strongest detector responses). This bounded keypoint processing limits the matching cost and stabilizes the step-level runtime on CPU platforms. When comparing different detectors and descriptors, the same constraint is applied to ensure controlled and fair evaluation under an identical keypoint budget, consistent with the evaluation protocol in Sections 4 and 5.

3.4. Feature Matching and Outlier Rejection

Descriptors extracted from I_k and I_{k+1} are matched to obtain tentative correspondence pairs. For floating-point descriptors (e.g., SIFT and SURF), Euclidean distance is commonly used, while Hamming distance is used for binary descriptors [37–39]. Approximate nearest-neighbor strategies may be employed to accelerate matching for high-dimensional descriptors, and descriptor-level filtering rules such as the ratio test are applied to suppress ambiguous correspondences [37,42]. The resulting tentative matches provide candidate correspondences for geometric model estimation.

Residual mismatches may remain due to repetitive patterns, low-texture regions, or motion blur. Robust geometric verification is therefore performed using random sample consensus (RANSAC) to separate inliers from outliers [35].

Candidate homographies are estimated from minimal correspondence subsets, and the model with the largest inlier support is selected under a reprojection-error threshold (in pixels). The final inlier set is then used for stable homography estimation. If inlier support does not satisfy a minimum acceptance criterion, the update can be rejected to prevent unstable alignment from propagating through sequential accumulation. For matching, we adopt conventional schemes (e.g., KNN with an appropriate nearest-neighbor search, such as FLANN for floating-point descriptors) without modifying the underlying matching algorithm, as the focus of this study is reliable CPU-feasible real-time operation and controlled evaluation under a fixed computational budget.

3.5. Homography Estimation and Accumulation

Inter-frame geometry between the reference frame I_k and the sequential frame I_{k+1} is modeled using a planar homography $\mathbf{H}_{k \rightarrow k+1}$. This projective model can represent typical UAV inter-frame transformations, including rotation, scale change, skew, and translation, and it is commonly adopted for incremental mosaicking when the scene is approximately planar or the dominant motion can be approximated as planar [15,19,21]. As illustrated in Figure 3, planar homography is estimated from inlier point correspondences between consecutive frames and then composed sequentially to map each frame into a common mosaic coordinate system.

Using homogeneous coordinates, a point $\tilde{\mathbf{x}}_{k+1}$ in the sequential frame and its corresponding point $\tilde{\mathbf{x}}_k$ in the reference frame satisfy Equation (4).

$$\tilde{\mathbf{x}}_k \sim \mathbf{H}_{k \rightarrow k+1} \tilde{\mathbf{x}}_{k+1} \quad (4)$$

where $\sim$ denotes equality up to scale. The homography is defined up to a global scale and can be written as Equation (5).

$$\mathbf{H} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \quad (5)$$

This mapping direction is consistent, where I_{k+1} is registered to I_k to support robust incremental updates under rotation, scale change, skew, and translation. The homography is estimated from the RANSAC inlier correspondence set using a linear method and then recomputed using all inliers after RANSAC selection to improve stability [37]. If the inlier support is below the acceptance criterion, the update can be rejected to avoid accumulating an unreliable transform in the global mapping.

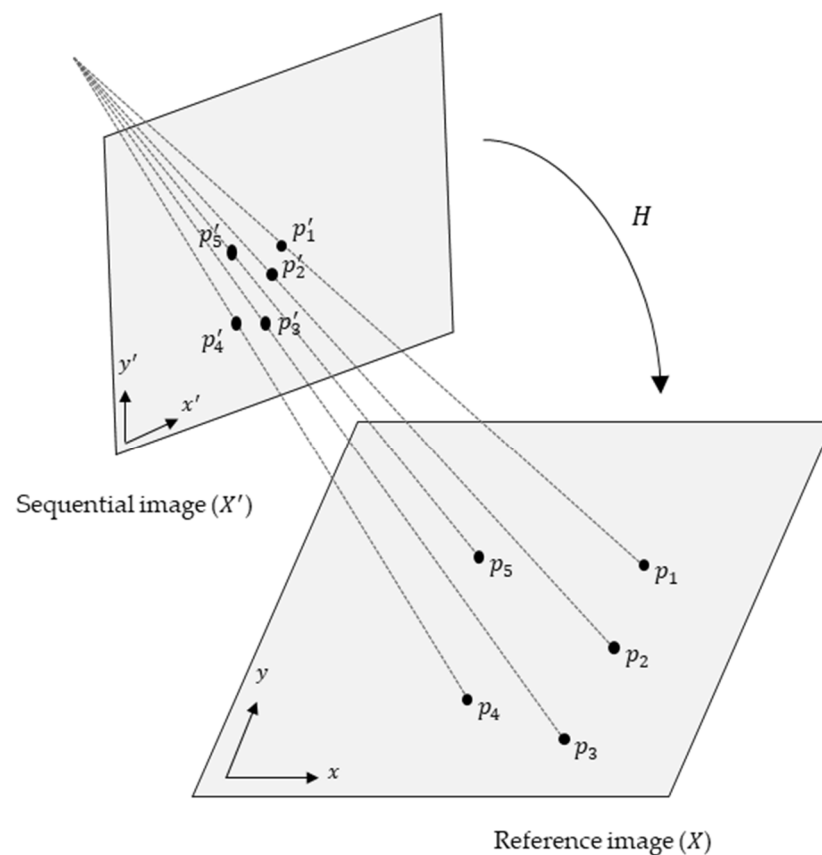


Figure 3. Planar homography estimation and sequential accumulation used for incremental mapping.

To map each extracted frame to a global mosaic coordinate system, inter-frame homographies are accumulated into a global transform $\mathbf{G}_k$ as Equation (6).

$$\mathbf{G}_0 = \mathbf{I}, \mathbf{G}_{k+1} = \mathbf{G}_k \mathbf{H}_{k \rightarrow k+1} \quad (6)$$

where $\mathbf{I}$ is the identity matrix. The accumulated transform maps the extracted frame at step t into mosaic canvas coordinates, enabling incremental mosaic growth with lightweight computation [15,19–21]. Because accumulation may introduce drift over long sequences, the operating regime emphasized in this work prioritizes stable short-horizon registration, sufficient overlap via sampling control, and conservative update acceptance.

The homography-based alignment adopted in this work assumes that the observed scene is approximately planar within the effective operating range of nadir-view UAV

imagery. Therefore, severe attitude changes and significant altitude variation may reduce registration stability due to increased parallax and non-planar effects, and such conditions are outside the intended operating scope of this study.

3.6. Incremental Orthomosaic Construction and Blending

The orthomosaic is represented on a global canvas in pixel coordinates. The first extracted frame is placed as initial content on the canvas. At update step t , the sequential frame is warped to the mosaic canvas using the accumulated transform G_k . Perspective warping and compositing are implemented using standard computer vision libraries that support homography-based transformation and image warping [43].

To maintain a valid canvas extent during mosaic growth, the four corners of the incoming frame are transformed by G_k , and an axis-aligned bounding box is computed in canvas coordinates. If the current canvas is insufficient to accommodate the warped frame, the canvas is expanded by padding while preserving the existing mosaic content. This strategy supports continuous mosaic growth during incremental mapping.

In overlap regions, a lightweight blending strategy is applied to maintain real-time throughput. Overwrite blending can be used as a baseline configuration, where pixel values from the latest warped frame replace existing values in overlap regions. This choice minimizes computational overhead and is suitable for CPU-oriented real-time visualization when registration is stable. Alternative blending strategies can reduce visible seams but increase computation. The incremental loop continues until all extracted frames are processed, resulting in the final 2D orthomosaic-style map [12–15,20–22].

Note that incremental homography composition may accumulate drift over long sequences, which is a known limitation of sequential alignment. Accordingly, this work focuses on a CPU-feasible real-time operating regime for nadir UAV video, emphasizing reliable short-horizon registration and continuous map updates rather than introducing additional global drift-mitigation steps.

4. Experimental Setup and Evaluation

This section describes the experimental setup and evaluation protocol used to characterize the proposed real-time orthomosaic-style mapping framework from UAV video. The evaluation is designed to clarify: (i) feature-extraction behavior under CPU-oriented constraints, (ii) matching and geometric registration stability under typical UAV motion, and (iii) the resulting mapping behavior and runtime-level real-time operability under consistent preprocessing and fixed parameter settings. This evaluation is intended as a controlled case study under fixed compute budgets, rather than a large-scale benchmark for statistical generalization across diverse video collections.

4.1. UAV Video Datasets and Test Scenarios

The evaluation uses UAV-captured RGB videos acquired during low-altitude flights over predominantly planar scenes. The sequences are recorded at 1920×1080 resolution and approximately 60 fps, which reflects a common operational setting for real-time monitoring and mapping. Representative frames from the test sequences are shown in Figures 4 and 5, illustrating visual characteristics that influence registration performance, including texture density and repetitive patterns.

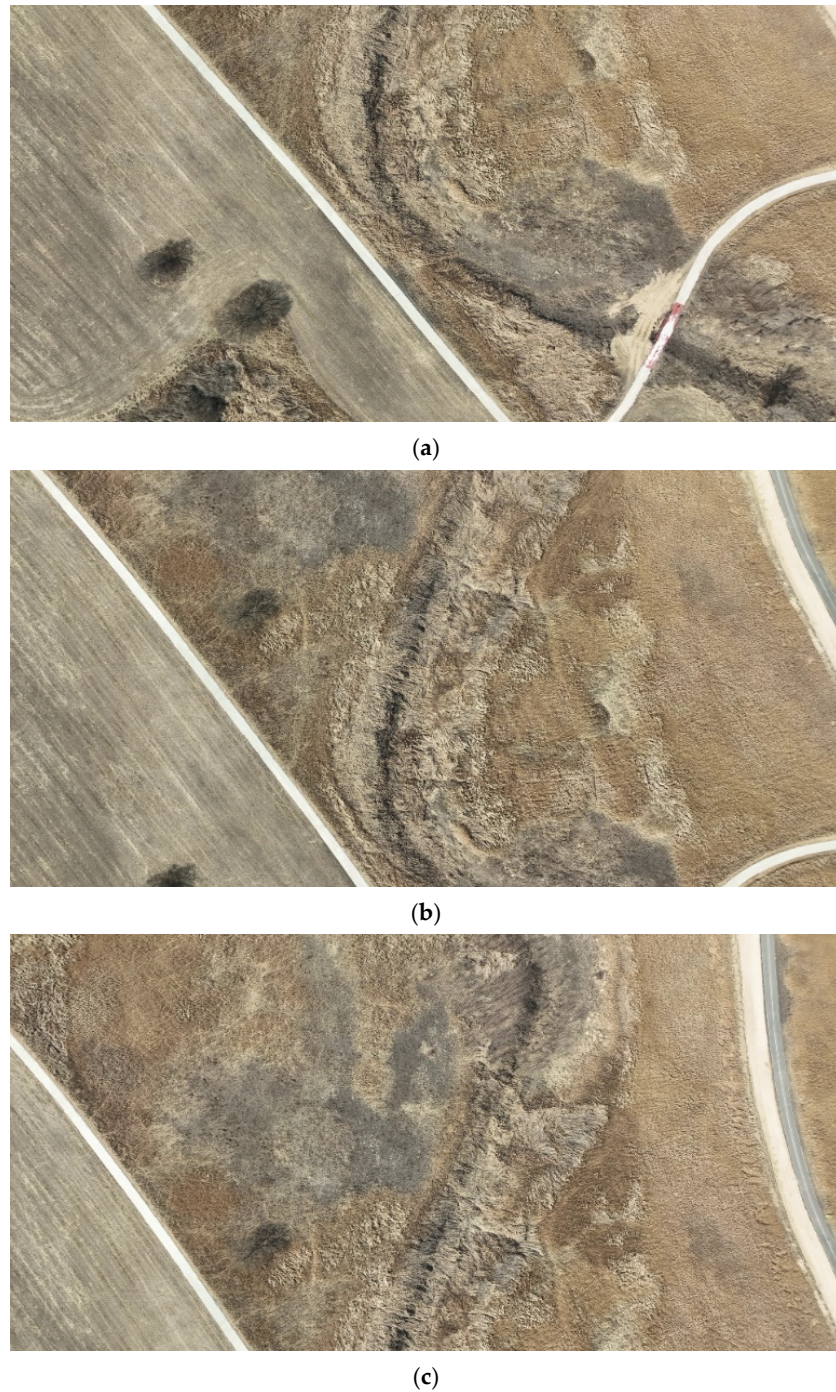


Figure 4. Nadir image Set 1 extracted from a drone-captured video at 60-frame intervals: (a) Frame 1; (b) Frame 2; (c) Frame 3.

These scene properties affect both the number and the spatial distribution of detected keypoints and consequently influence the robustness of homography estimation in an incremental mapping procedure.

To provide an interpretable comparison of feature and matching behaviors, two representative frames are selected from the datasets and are referred to as Image 1 and Image 2 throughout this section. The corresponding RGB-to-grayscale conversion outcomes are shown in Figure 6 to highlight differences in texture and contrast that directly impact detector responses and descriptor matching.

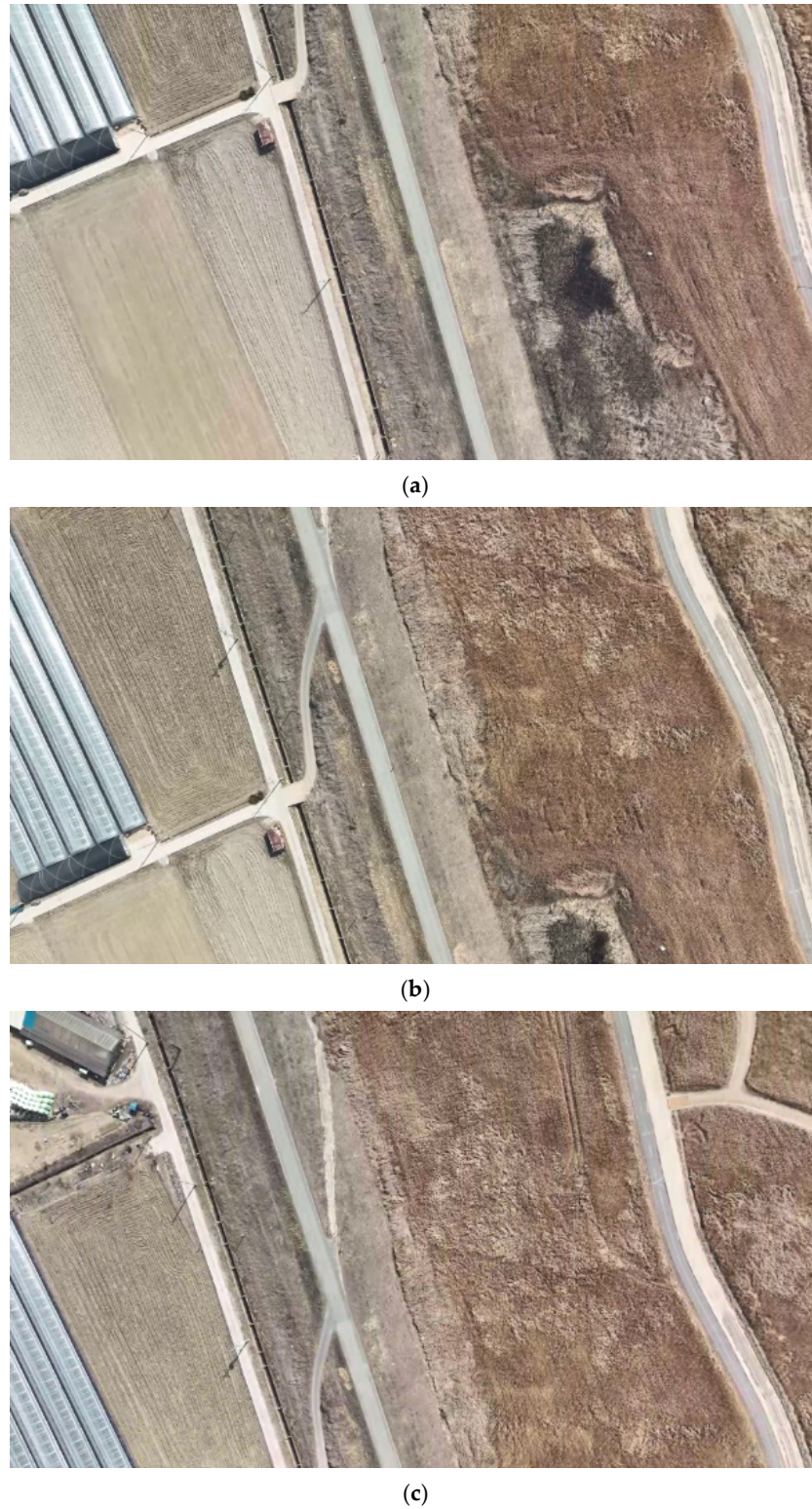


Figure 5. Nadir image Set 2 extracted from a drone-captured video at 60-frame intervals: (a) Frame 1; (b) Frame 2; (c) Frame 3.

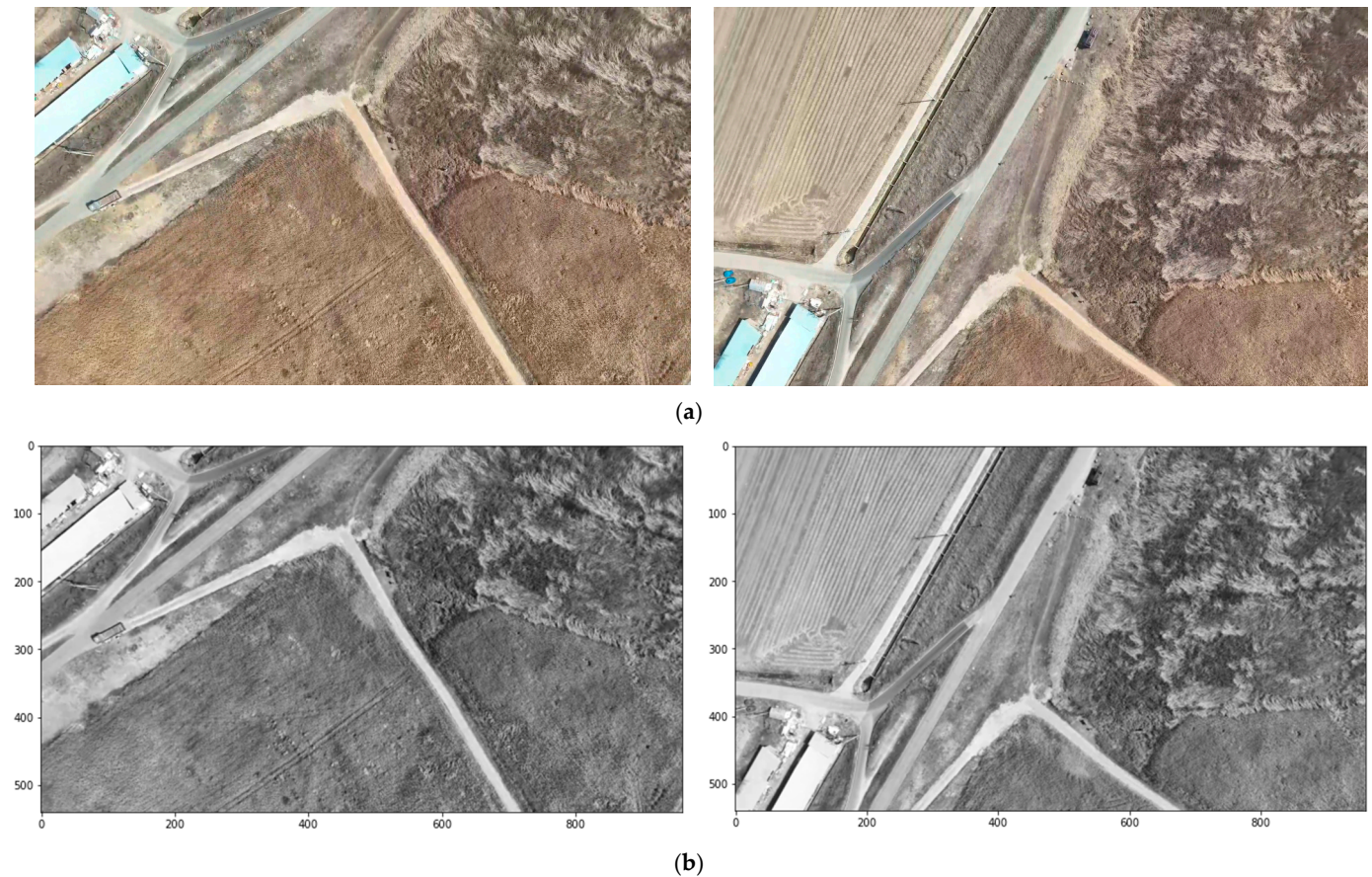


Figure 6. Drone-captured images for feature point extraction: (a) original; (b) gray-scale images.

4.2. Preprocessing and Sampling Settings

All extracted frames are processed using a consistent preprocessing pipeline prior to feature extraction. Each RGB frame is converted to a grayscale image to reduce computational load and to stabilize feature detection by operating on intensity structures. Grayscale conversion preserves luminance-driven gradients and corner responses that dominate classical detectors, while reducing chromatic variations that may introduce non-essential variability under changing illumination.

Frame sampling is applied to balance overlap and throughput. Rather than processing every video frame, frames are extracted at a fixed interval (every k -th frame). The visual impact of the sampling choice on inter-frame overlap and apparent motion is illustrated in Figures 4 and 5, where adjacent extracted frames exhibit viewpoint and scale differences typical of UAV video.

When reporting real-time operability, the sampling interval is interpreted as an update period $\Delta t = k/f$, where f is the video frame rate. Optional resizing and cropping are applied only when needed to maintain stable runtime and to suppress non-ground regions. If resizing is used, the same resizing rule is applied to all frames within each sequence. If cropping is used, a fixed crop region is applied consistently to avoid introducing time-varying distortions. The qualitative appearance of extracted keypoints and their spatial distribution under different configurations are visualized in Figure 7. These differences are particularly relevant for UAV imagery because registration stability can be sensitive to whether keypoints are concentrated in high-texture regions or distributed more uniformly across the field of view.

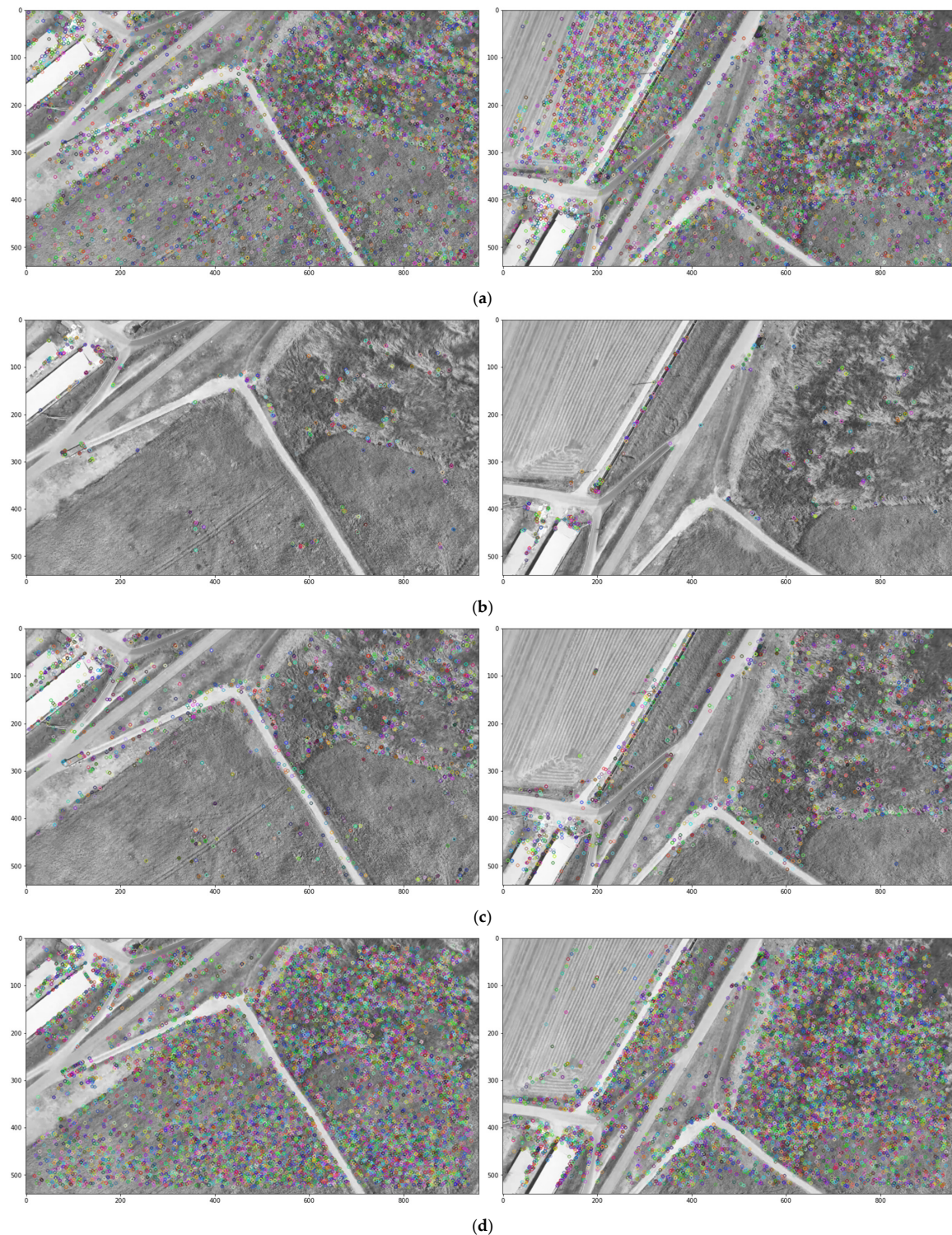


Figure 7. Extracted feature points from the drone images: (a) SIFT; (b) ORB; (c) KAZE; (d) BRISK.

4.3. Feature-Extraction Configurations

Feature extraction affects both registration robustness and computational feasibility. Classical local features are evaluated to reflect practical deployment scenarios in which training-free and CPU-feasible methods are preferred. In this study, SIFT, ORB, KAZE, and BRISK are considered as representative detectors/descriptors available in standard computer vision libraries.

The principal detector and descriptor settings, including keypoint selection criteria and descriptor representation, are summarized in Table 1. Parameters are fixed for each method to enable consistent comparison across datasets. For fair comparison and predictable runtime, a fixed keypoint budget is applied in the registration pipeline. Except for ORB (internally capped via the OpenCV parameter *nFeatures*), only the strongest keypoints are retained up to *N* for subsequent matching and homography estimation.

Table 1. Feature-extraction configurations used in the experiments, including detector and descriptor settings.

Method	OpenCV Parameter	Descriptor Dimension, Type	Descriptor Memory
SIFT	nOctaveLayers = 3, contrastThreshold = 0.04, edgeThreshold = 10, sigma = 1.6	128 d, float32	512 bytes
ORB	nFeatures = 500, scaleFactor = 1.2, nLevels = 8, edgeThreshold = 31, WTA_K = 2, patchSize = 31, fastThreshold = 20	256-bit, binary	32 bytes
KAZE	extended: false (64), upright: false, threshold: 0.001, <i>n</i> octaves: 4, <i>n</i> octave layers: 4, diffusivity: 1	64 d, float32	256 bytes
BRISK	thresh = 30, octaves = 3, patternScale = 1.0	512-bit, binary	64 bytes

Practical feasibility is assessed by measuring (i) the number of detected keypoints (before applying $N_{\max}$) and (ii) the feature-extraction time under identical grayscale pre-processing and sampling conditions. The resulting statistics for Image 1 and Image 2 are summarized in Table 2.

Table 2. Numbers of detected and retained keypoints and feature-extraction time measured on two representative grayscale frames.

Method	Image 1			Image 2		
	Detected Keypoints	Used Keypoints	Time [s]	Detected Keypoints	Used Keypoints	Time [s]
SIFT	4792	500	0.06	6682	500	0.07
ORB	500	500	0.01	500	500	0.01
KAZE	1382	500	0.18	1734	500	0.18
BRISK	10,670	500	0.11	8993	500	0.10

4.4. Matching Strategy Configurations

After feature extraction, descriptor matching is performed to obtain tentative correspondences between consecutive grayscale frames I_t and I_{t+1} . Matching strategies are compared to assess their impact on both runtime and correspondence quality under a controlled keypoint budget.

To isolate the effect of the matching strategy, Table 3 compares matching configurations under fixed feature-extraction settings and an identical acceptance rule. Tentative correspondences are filtered using a fixed ratio-test threshold prior to geometric verification. The visual characteristics of tentative correspondences under UAV motion are illustrated in Figure 8. These results support the interpretation of Table 3 in terms of both geometric support and computational time.

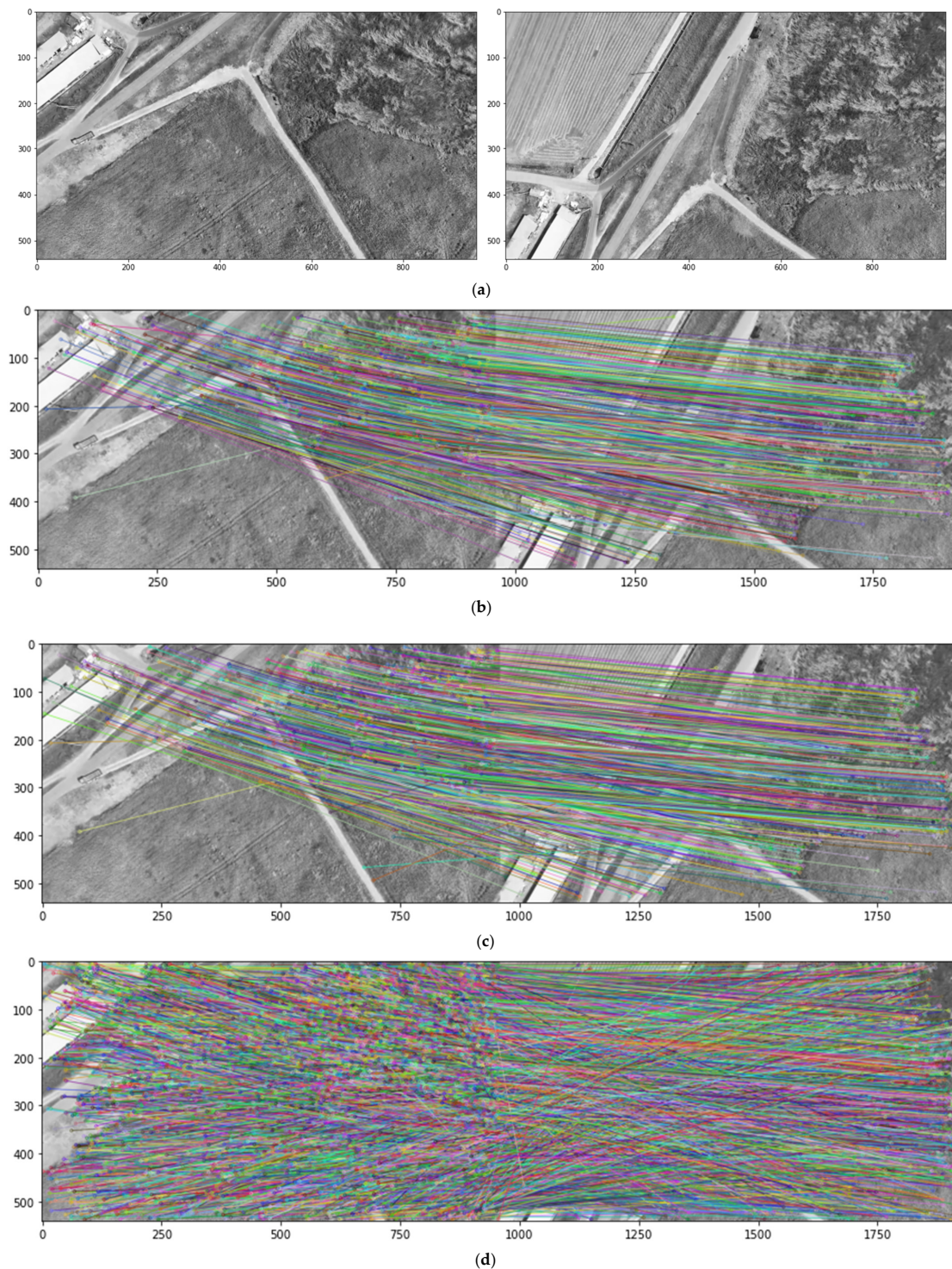


Figure 8. Matching results by algorithm: (a) Gray-scale image; (b) fast library for approximate nearest neighbors; (c) k -nearest neighbors (KNN); (d) brute force (BF).

Table 3. Comparison of descriptor matching methods under fixed feature extraction and identical ratio-test filtering.

Method	Matching Points [EA]	Avg. Distance	Processing Time [s]
FLANN	500	151.748	0.056
KNN	500	157.272	0.059
BF	500	272.515	0.124

To support reproducibility and fair comparison, all configurations are evaluated under an identical frame sampling rule, a fixed keypoint budget, and consistent matching and RANSAC-based geometric verification criteria. Unless stated otherwise, the processing is performed on the CPU without GPU acceleration, and all timing values correspond to wall-clock measurements under the same execution environment.

4.5. Geometric Verification and Homography-Based Registration Settings

Geometric verification is performed using a robust estimator to reject outliers and to obtain a stable set of inlier correspondences for homography estimation. A reprojection-error threshold is used to determine inlier membership, and an iteration budget is selected to maintain stable performance under typical UAV motion.

The practical effect of outlier rejection and the resulting inlier structure are illustrated in Figure 9, which compares tentative correspondences before RANSAC with inlier correspondences after RANSAC. Homography estimation follows the projective model, supporting rotation, scale change, skew, and translation observed in UAV video.

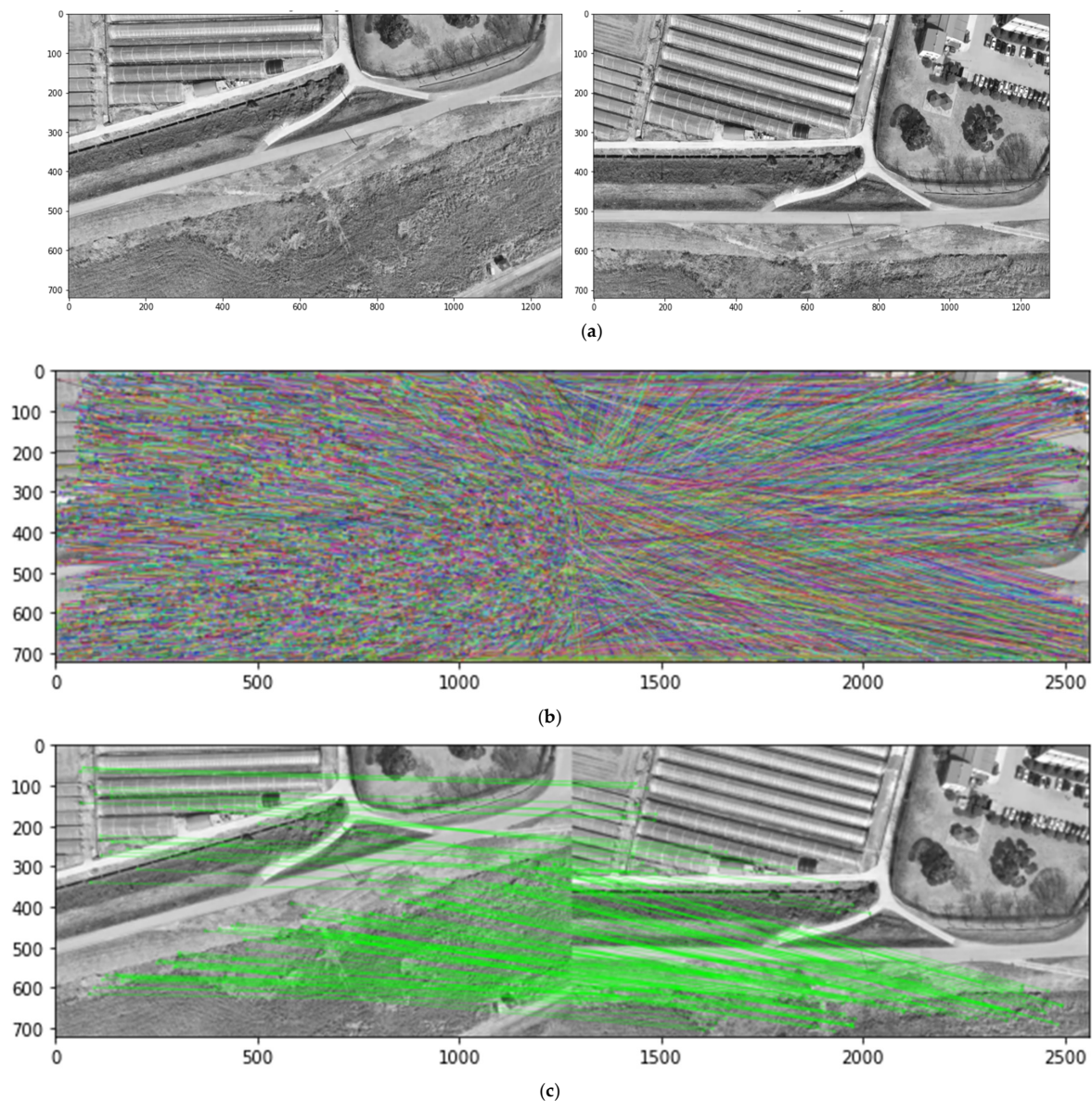


Figure 9. Results of applying the random sample consensus (RANSAC) algorithm: (a) Gray-scale images; (b) before RANSAC application; (c) after RANSAC application.

To ensure consistent evaluation, the same RANSAC inlier criterion and the same minimum inlier requirement are applied across all feature/matching configurations. This prevents configuration-dependent tuning and enables direct comparison of geometric consistency under identical verification rules.

4.6. Pairwise Registration Assessment

Prior to full orthomosaic construction, pairwise registration quality is assessed to confirm that the selected configuration yields stable alignment between consecutive frames under representative UAV motion. This assessment focuses on whether the estimated projective transform produces coherent alignment over the overlap region and whether misregistrations or local distortions are evident at the pairwise level.

Representative pairwise registration results are shown in Figure 10 at the native image resolution. These examples provide a sanity check that the adopted configuration can align consecutive frames across different textures and imaging conditions, supporting its use in sequential homography accumulation and incremental mosaic updates.

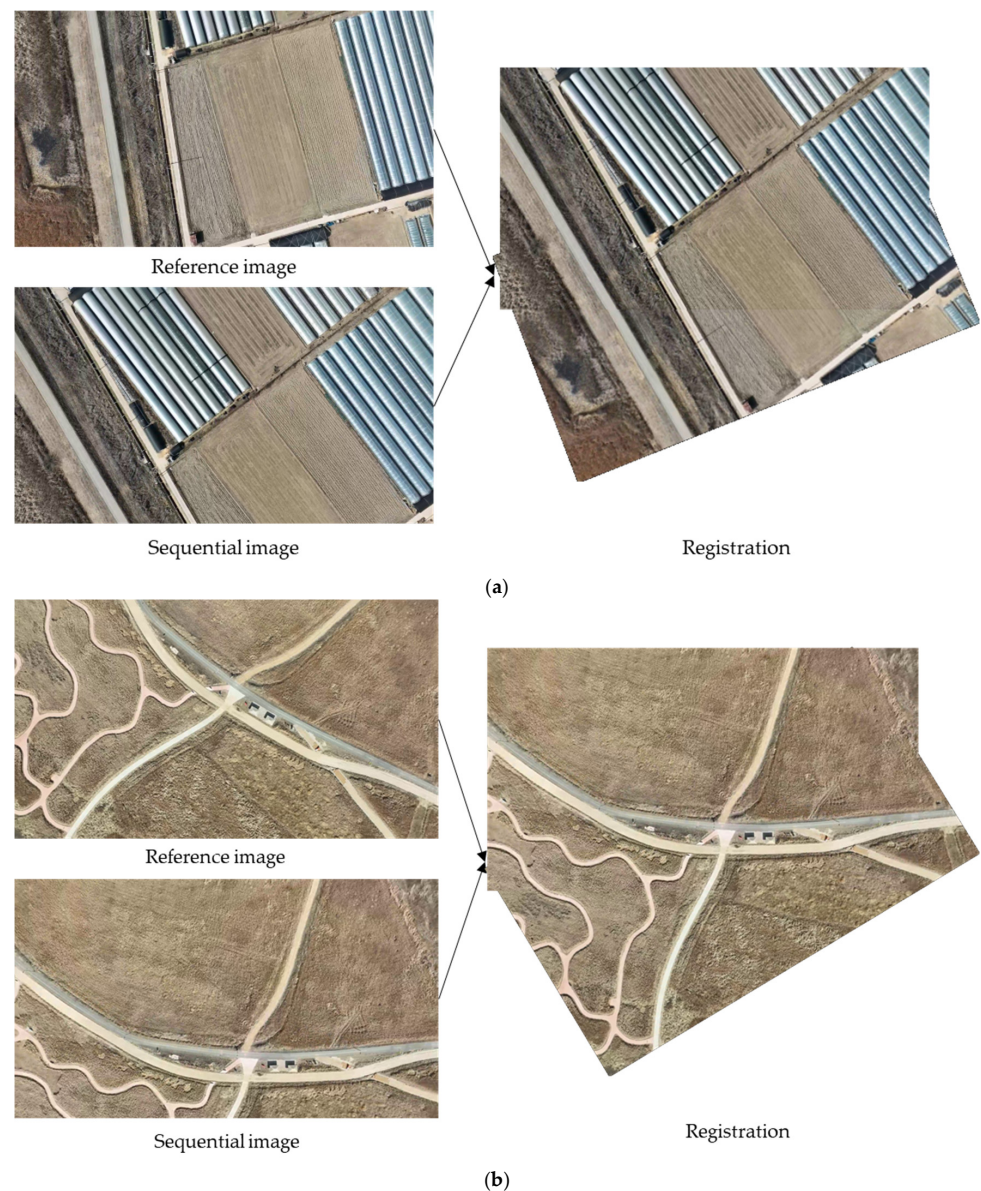


Figure 10. Pairwise registration examples of the captured images (1920×1080 pixels): (a) Example 1; (b) Example 2.

4.7. Ablation Study Design for Real-Time Parameters

Real-time operation in the proposed pipeline is primarily governed by two controllable design factors: the frame sampling interval k and the keypoint budget N . Increasing k reduces the number of processed frames per unit time but may decrease inter-frame overlap and increase the risk of registration failure when camera motion is large. Increasing N generally improves the chance of obtaining sufficient inlier correspondences for homography estimation, but increases computational cost due to descriptor extraction and matching.

To discuss this trade-off in a structured manner, the evaluation records the following runtime- and alignment-related indicators under fixed preprocessing, matching rules, and RANSAC settings: (i) per-step processing time, (ii) the real-time feasibility condition with respect to $\Delta t = k/f$, and (iii) match statistics such as the number of tentative correspondences and the RANSAC inlier ratio. These indicators are used in Section 5 to support an objective interpretation of mapping behavior and CPU feasibility under different operating points.

The ablation discussion is presented as a practical interpretation rather than an exhaustive statistical study. The focus is placed on identifying parameter sensitivities and failure modes observed in the tested sequences, and on explaining how conservative parameter choices can improve stability while maintaining the feasible speed. These observations should be interpreted as configuration guidance under the tested conditions, not as a claim of broad statistical effectiveness.

5. Results and Practical Interpretation

This section reports results obtained under the fixed evaluation protocol described in Section 4. The results are organized to present (i) orthomosaic-style mapping outcomes produced by incremental homography-based updates, (ii) CPU runtime behavior under the fixed settings, and (iii) a practical discussion of how key real-time design parameters (the frame sampling interval and the keypoint budget) influence update stability and runtime characteristics. Unless stated otherwise, all comparisons follow identical preprocessing, matching, and RANSAC verification criteria to ensure that observed differences reflect the tested design parameters rather than configuration-dependent tuning.

This evaluation is intended as a case-based demonstration of operational behavior under fixed CPU constraints, and it does not establish statistical generalization across diverse UAV video conditions.

5.1. Orthomosaic Mapping

This section presents orthomosaic-style mapping results produced by the proposed incremental registration and mosaic update pipeline on three nadir UAV video sequences. Figures 11–13 and Table 4 show representative sampled frames from Drone Videos 1–3, respectively, and Figures 14–16 show the corresponding orthomosaic outputs generated by sequential homography estimation and incremental compositing. These examples are intended to illustrate typical mapping behavior and common degradation patterns under the fixed operating settings, rather than to provide statistically comprehensive validation across broad scene categories.

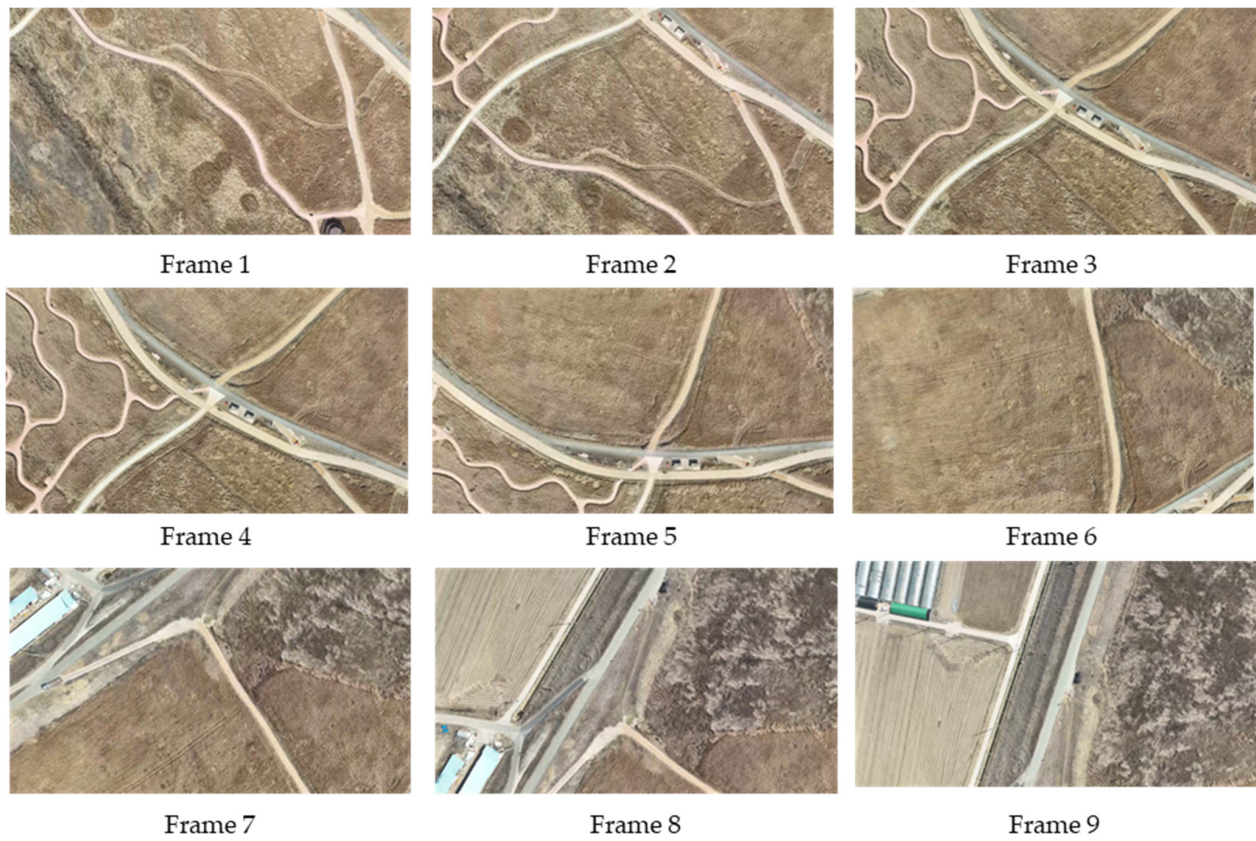


Figure 11. Extracted frame-by-frame images from Drone Video 1.



Figure 12. Extracted frame-by-frame images from Drone Video 2.

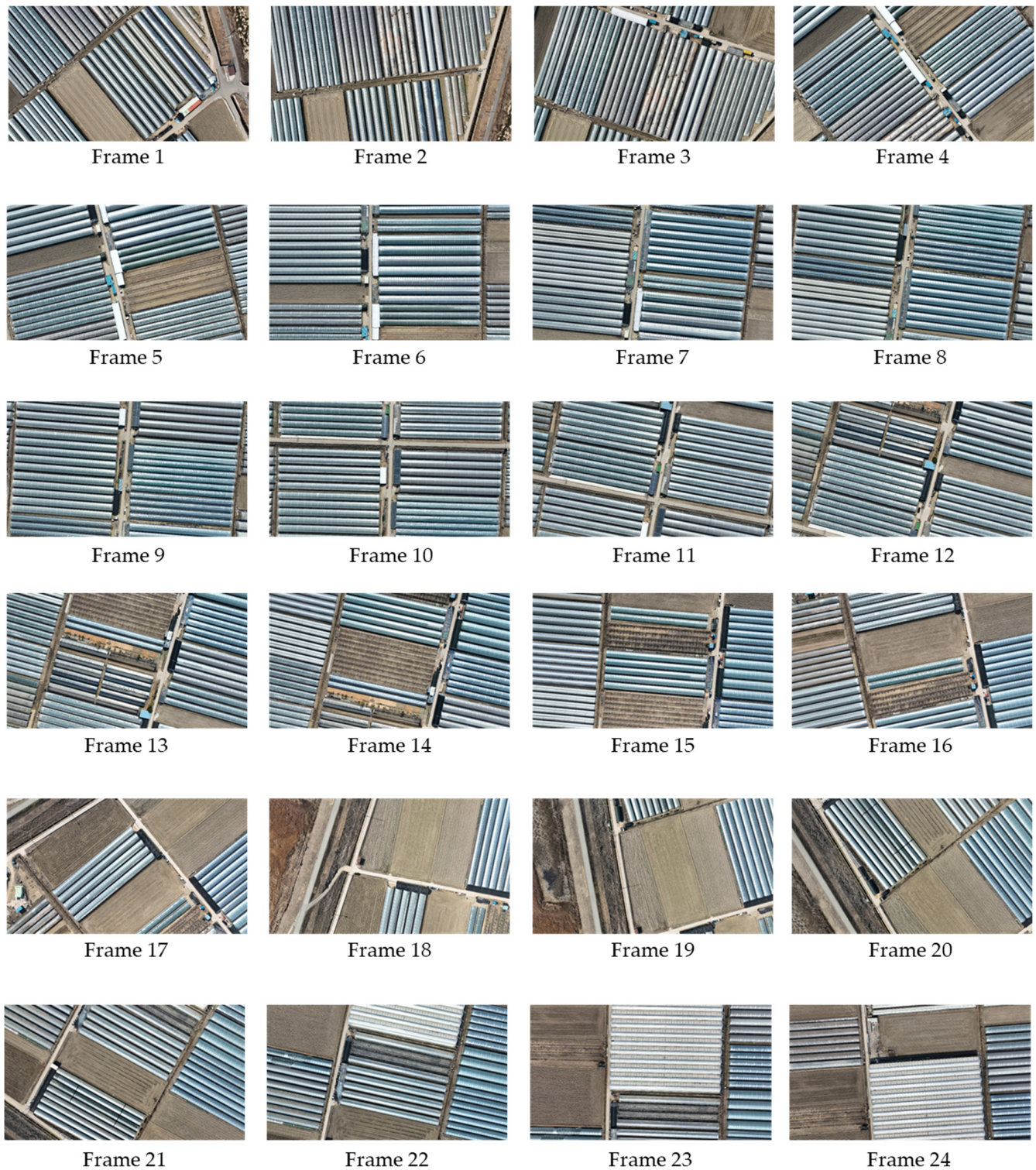


Figure 13. Extracted frame-by-frame images from Drone Video 3.

Table 4. Drone videos for 2D orthomosaic map generation.

	Size [Pixels]	Recording Time [s]	Flight Path
Drone Video 1	1920 × 1080	62	Circling
Drone Video 2	1920 × 1080	55	Circling
Drone Video 3	1920 × 1080	160	Circling, straight

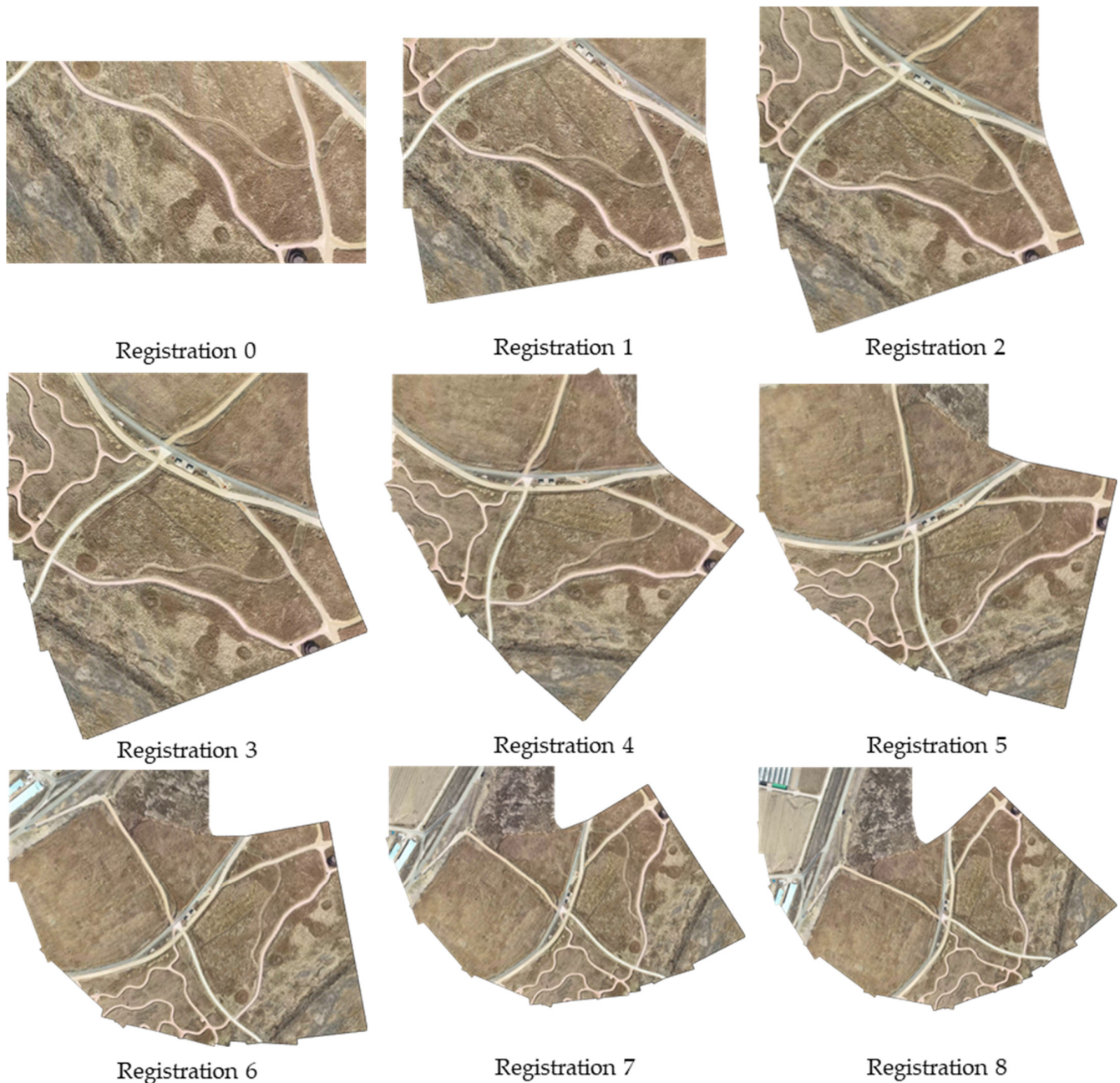


Figure 14. Sequentially registered 2D orthomosaic map image of Drone Video 1.

Drone Video 1 (Figures 11 and 14). Figure 11 shows sampled frames that contain road markings and surrounding textures, which typically provide stable local keypoints. Under these visual characteristics, frame-to-mosaic registration tends to yield consistent correspondences across successive updates. The resulting orthomosaic in Figure 14 forms a continuous 2D overview with visually coherent alignment over most regions. Minor seam visibility may appear where local contrast changes or perspective effects increase; however, the overall layout remains stable under the approximately planar scene assumption.

Drone Video 2 (Figures 12 and 15). Figure 12 shows a scene dominated by man-made structures and repeated patterns (e.g., building edges and similar roof textures), which can yield many candidate keypoints but may also introduce ambiguous correspondences. Under these conditions, the registration quality depends on the distribution of reliable inliers after robust estimation. The orthomosaic in Figure 15 shows that the proposed pipeline can integrate a wide field of view into a single map, while local seam artifacts or

slight distortions can become more noticeable in areas with repeated structures or weakly textured regions. These behaviors are consistent with the known sensitivity of feature matching under repetitive patterns, even when robust estimation is applied.

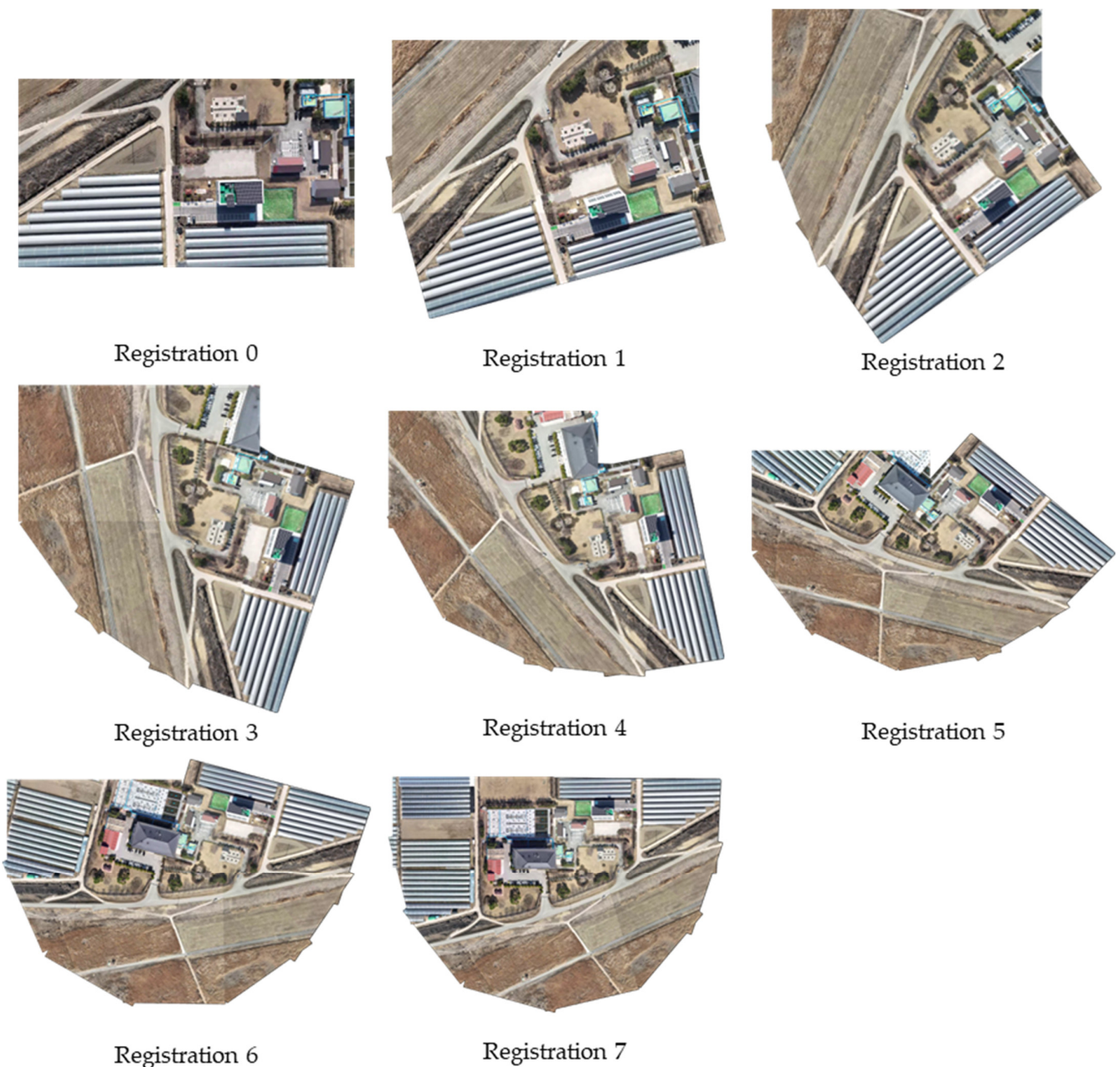


Figure 15. Sequentially registered 2D orthomosaic map image of Drone Video 2.

Drone Video 3 (Figures 13 and 16). Figure 13 shows sampled frames from a longer sequence with mixed motion characteristics (straight and circling segments), which introduces larger appearance changes and a wider range of viewpoint variation compared with the other sequences. The orthomosaic in Figure 16 remains visually consistent over large portions of the covered area when sufficient overlap and discriminative texture are present. At the same time, localized misalignment or seam visibility may appear in portions affected by motion blur, low-texture surfaces, or non-planar structures that violate the planar homography assumption. This example is included to illustrate how incremental homography accumulation behaves over a longer trajectory under the same fixed settings, and to make typical failure/degradation modes explicit.

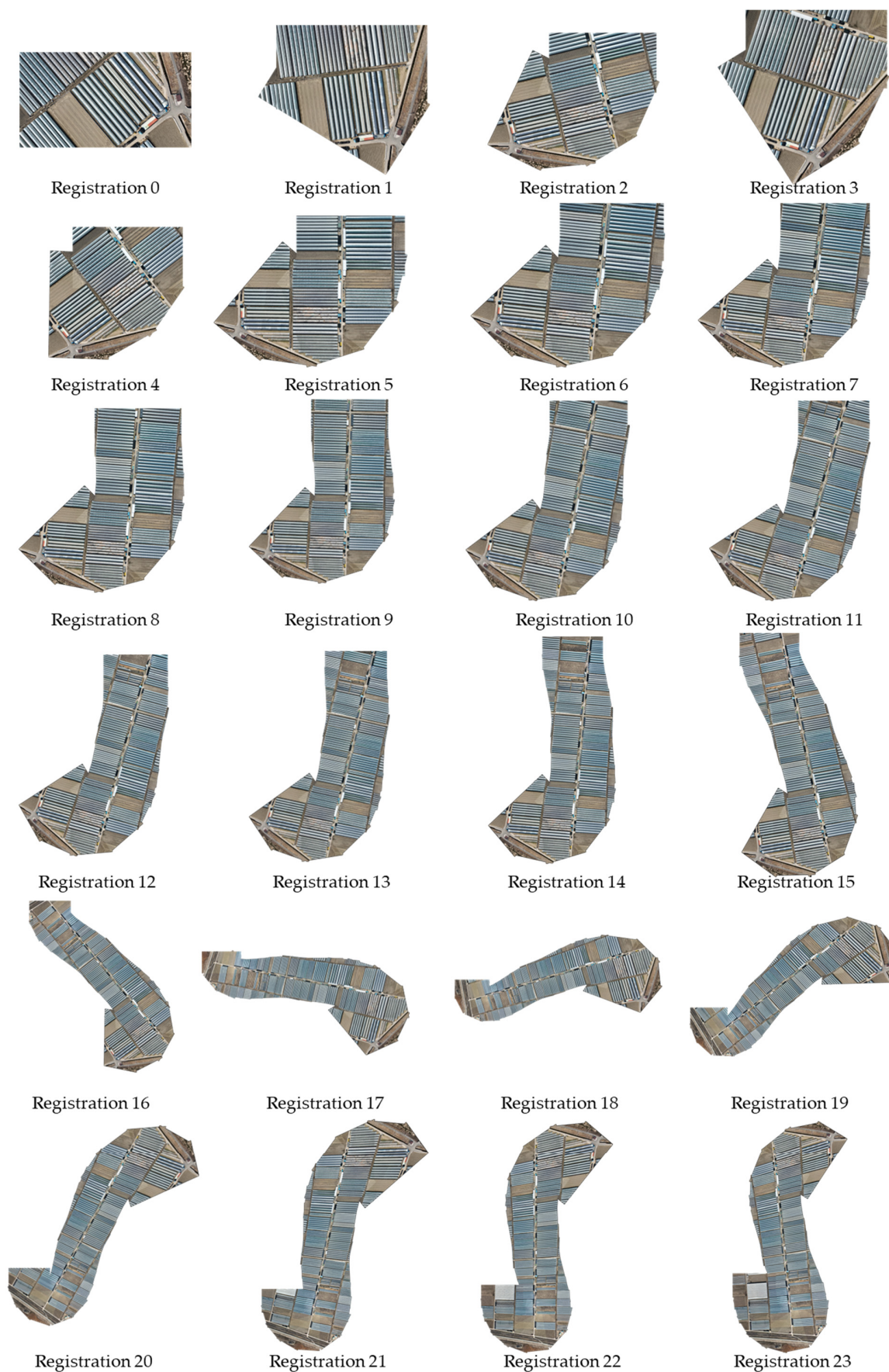


Figure 16. Sequentially registered 2D orthomosaic map image of Drone Video 3.

The evaluation uses three UAV video sequences with different recording durations and flight patterns. A summary of the datasets is provided in Table 4, and representative sampled frames are used to illustrate inter-frame overlap and scene characteristics relevant to registration. We emphasize that the three sequences are used to report representative behavior under controlled settings, and that broader validation on a substantially larger number of sequences is left for future benchmarking. Therefore, the presented results should be discussed as illustrative case studies rather than statistically comprehensive validation.

5.2. Real-Time Feasibility

This section reports CPU runtime measurements of the proposed incremental orthomosaic-style mapping pipeline under the fixed experimental settings. The runtime is evaluated at the registration–update-step level, where each update includes feature extraction, descriptor matching, robust homography estimation, and mosaic update. For each UAV sequence, Table 5 summarizes the recording time and the number of extracted frames used for incremental updates, together with the total elapsed processing time and per-step CPU time statistics (mean, 95th percentile (P95), and maximum). Table 5 shows that the per-step CPU time remains on the sub-second scale across the tested sequences under the selected configuration. The mean step time reflects typical processing behavior, whereas the P95 and maximum values describe upper-tail behavior and occasional slower updates. The gap between the mean and the maximum values indicates step-to-step variability, which may arise from scene-dependent correspondence quality and the amount of robust estimation required to obtain a valid homography in a given update.

Table 5. Registration update counts and CPU time per registration step for each video sequence.

	Recording Time [s]	Extracted Frames	Total Elapsed Time [s]	Mean Per-Step CPU Time [s]	95th Percentile [s]	Max [s]
Drone Video 1	62	9	3.160	0.395	0.571	0.594
Drone Video 2	55	8	2.986	0.354	0.524	0.524
Drone Video 3	160	24	6.309	0.274	0.410	0.562

The total elapsed time provides an end-to-end view of the processing cost for each sequence under the same setting. Since the total elapsed time scales with the number of extracted frames (updates), it is consistent with the use of per-step statistics to characterize computational behavior. These measurements are reported to document the practical runtime characteristics of the pipeline on CPU for the tested sequences; broader claims about runtime performance across diverse acquisition conditions require larger-scale evaluation. To connect these runtime observations to design choices, the next section (Section 5.3) discusses key parameters that directly affect per-step computation, including the sampling interval (i.e., how frequently frames are processed) and the keypoint budget. Rather than changing the pipeline structure, the discussion isolates how these parameters influence the expected update frequency and the robustness of geometric estimation under CPU constraints, providing practical guidance for selecting operating points.

5.3. Discussion on Design Parameters for Real-Time Deployment

This section discusses two practical parameters that govern the behavior of the proposed incremental orthomosaic-style mapping pipeline under CPU constraints: the sampling interval (processed every k -th frame) and the keypoint budget (upper bound on the number of local features per update). These parameters do not alter the pipeline structure, but they directly affect update responsiveness, runtime predictability, and the stability of planar homography estimation.

A smaller sampling interval generally increases inter-frame overlap and reduces viewpoint change between processed frames, which tends to strengthen correspondence reliability and stabilize homography estimation. In contrast, a larger sampling interval can increase appearance change and reduce overlap, making the registration more sensitive to weak texture, motion blur, repeated patterns, and parallax effects that violate the planar assumption. The keypoint budget controls the per-step computational load: increasing $N_{\max}$ can improve robustness in challenging scenes by providing more candidate correspondences, but it can also increase matching and robust estimation cost and may exhibit diminishing returns beyond a moderate budget.

For real-time operation, mean per-step runtime alone is often insufficient because occasional slow updates can cause visible latency. Practical CPU operation should therefore be interpreted using conservative runtime behavior (e.g., upper-tail statistics) relative to the update period $\Delta t = k/f$, ensuring that per-step processing remains reliably below Δt with an adequate buffer. Overall, k primarily governs overlap-driven geometric stability and update responsiveness, whereas $N_{\max}$ primarily governs computational predictability and correspondence robustness, and appropriate combinations of these parameters define a practical operating regime for CPU-only incremental mapping.

Direct one-to-one comparisons with recent lightweight real-time mosaicking/registration pipelines are challenging due to differing assumptions, implementation availability, and evaluation protocols across methods. In this study, we therefore restrict the discussion to (i) controlled comparisons within the proposed pipeline under fixed compute budgets and (ii) a qualitative contrast to offline SfM/MVS workflows in terms of deployment characteristics (offline processing versus incremental CPU updates). A broader benchmark across a larger number of diverse video sequences and alternative real-time baselines under a unified protocol is left for future work.

6. Conclusions

This paper presented a CPU-feasible, real-time incremental framework for generating 2D orthomosaic-style maps directly from nadir UAV video. The proposed pipeline performs frame-by-frame registration using classical local features, robust matching with RANSAC-based geometric verification, and a full planar homography model to represent typical inter-frame transformations (rotation, scale change, skew, and translation). Real-time operation is supported through explicit system-level controls, including fixed frame sampling and a bounded keypoint budget, enabling continuous mosaic updates without relying on offline SfM/MVS reconstruction.

Results on three UAV video sequences illustrate typical mapping behavior under the fixed operating settings. When sufficient overlap and discriminative texture are available and the scene is reasonably approximated as locally planar, the proposed pipeline produces visually coherent orthomosaic-style overviews with consistent incremental updates. CPU runtime measurements further document per-step processing-time characteristics (mean, P95, and maximum) under the selected configuration, providing practical evidence of predictable runtime behavior for the tested sequences.

Several limitations should be noted. First, the proposed outputs are orthomosaic-style 2D visualizations and are not intended to replace metrically accurate, georeferenced orthomosaics produced by full photogrammetric pipelines when high-precision mapping is required. Second, homography-based incremental registration is inherently sensitive to violations of the planar (or locally planar) assumption; scenes with strong parallax, complex 3D structure, severe attitude changes, altitude variations, or weak texture can degrade alignment stability and increase visible seam artifacts. Third, drift may accumulate as sequence length increases due to sequential transform composition; this work does

not incorporate global optimization, loop closure, or sensor fusion modules that can mitigate drift but typically increase system complexity and computational cost. Fourth, the evaluation scale is limited; while the reported results document typical behavior and runtime characteristics under controlled compute constraints, broader generalization across diverse scenes, durations, and motion conditions requires larger-scale benchmarking with additional video sequences. Finally, precise quantitative accuracy metrics (e.g., absolute registration error) would require ground-truth references such as surveyed control points or georeferenced maps, which were not available for the tested videos.

Future work will focus on extending evaluation to a larger and more diverse set of UAV videos, and on incorporating lightweight stabilization strategies that reduce drift while maintaining CPU feasibility. Additional comparisons under a unified protocol with representative real-time baselines, as well as optional integration of navigation priors when available, are also promising directions for practical deployment.

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